HARDWARE DESIGN BRIEF

LoRa Soil Sensor Node

PRODUCT	LoRa soil sensor
RADIOS	868 MHz LoRa · 2.4 GHz BLE
ANTENNAS	868 MHz PCB · 2.4 GHz in module
POWER	USB-C or external 2×AA holder
PCB	4 layers · 0.8 mm finished
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DESIGN SCOPE

Function	Battery-capable soil sensor node with an 868 MHz LoRa link.
Controller	ESP32-S3-WROOM-1 module with its integrated 2.4 GHz antenna.
Sensor	External Adafruit STEMMA soil sensor connected by 3.3 V I^{2}C.
Power	USB-C or two AA cells in series through an external holder. No charger.
Power cycle	PCB-mounted 15-minute timer switches the ESP32, LoRa module and sensor off together.
Connectors	USB-C, battery and I^{2}C connectors are fitted. Unused connectors remain empty.

SYSTEM DESCRIPTION

U6 reads the external soil sensor over I^{2}C and communicates with U5 over SPI. U5 drives the 868 MHz antenna through a 50 Ω feed and matching network. The ESP32 module provides the 2.4 GHz antenna. J1 provides USB power and USB D+/D−; J3 connects to an external 2×AA holder.

U4 starts the switched 3.3 V rail every 15 minutes. A DONE edge from IO4 switches the rail off. U6, U5 and J2 are powered only from this rail and are unpowered between cycles. Only the signals listed as firmware-fixed on sheet 5 have mandatory GPIO numbers; the other GPIO numbers are recommendations.

ARRANGEMENT

top view · not a fixed outline

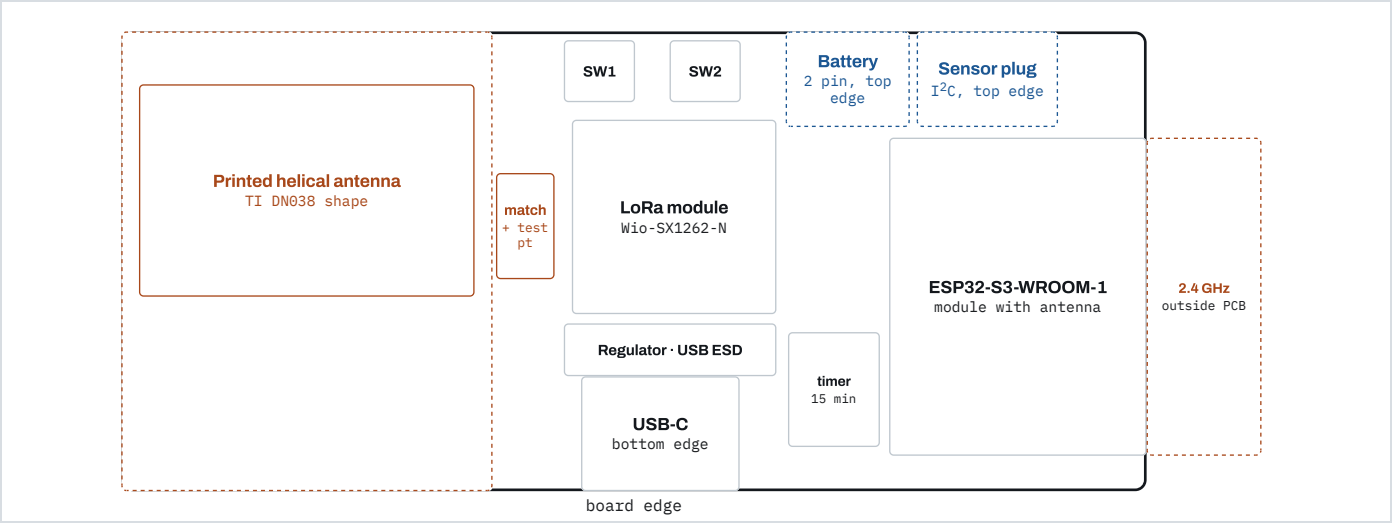


Fig 1 — Conceptual arrangement of the main parts, seen from the top. This is not a dimensioned placement drawing. Use the manufacturer courtyards and the antenna geometry on sheet 2. One radio sits at each end with the ground plane between them. The antenna section of the ESP32 module must sit completely outside the PCB outline. The battery and sensor plugs (blue) sit on the top edge, away from the USB-C cable. The timer sits beside the power converter in the grounded centre of the board. Small resistors and capacitors fill the free space.

Antennas and Manufacturing

LORA	868 MHz printed helical
BLUETOOTH	ESP32 module antenna
PCB	4 layers, minimum practical size
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ANTENNAS

main-PCB LoRa antenna · certified ESP32 module antenna

868 MHZ · LORA

Printed helical antenna

DESIGN	Import the TI DN038 (SWRA416) 868 MHz Gerber antenna and its feed/ground boundary without redrawing. Do not reduce the adjacent ground area shown by TI. Extend the main-board ground only away from the antenna, then tune on the finished board in its enclosure
COPPER	**19 × 12 mm**, on the top and bottom layers, joined by vias
CLEAR AREA	Only the DN038 top/bottom helix copper and its inter-layer vias may enter the antenna area. Those vias are part of the antenna and must never connect to GND. Keep all planes, tracks, parts and unrelated copper out of the Gerber-defined antenna area on every layer
MATCHING	Pi network (one series part, two shunt parts) at the feed point. TI used a 1.0 pF series capacitor and a 12 nH shunt inductor
FEED AND TEST	Use a 50 Ω line calculated from the released stack-up. Connect U5 ANT → R18 (0 Ω) → the radio-side matching node. Connect J4 centre → R19 (0 Ω) → the same node; J4 shell goes to a dense ground return. Production: R18 fitted, R19 and J4 DNP. VNA setup: remove R18, then fit R19 and J4. Keep the DNP branch short and record the VNA reference plane

2.4 GHZ · BLUETOOTH AND WI-FI

Antenna inside the module

DESIGN	Use the antenna integrated into the ESP32-S3-WROOM-1 module. Do not add an external 2.4 GHz matching network or antenna
COMPLIANCE	Follow the module manufacturer's antenna placement and enclosure conditions
PLACEMENT	Place the antenna end of the module **completely outside the PCB outline**
ON THE PCB	Follow Espressif's module keep-out drawing. This project requires the antenna section outside the base-board outline, with ground copper and stitching vias ending at the specified boundary
ENCLOSURE	At least **15 mm of free space in every direction** around the antenna once the board is in its case. See the Espressif layout guidelines

PCB AND MANUFACTURING REQUIREMENTS

LAYOUT

- **Use the minimum practical outline** that satisfies component courtyards, both antenna keep-outs and the DN038 feed/ground geometry. Do not reduce the DN038 adjacent ground area.
- **Four layers, 0.8 mm finished thickness.** Obtain the fabricator's released impedance stack-up before routing. Signals are on top and bottom; one inner layer is ground and the other is power. Remove both inner planes from the defined 868 MHz and 2.4 GHz antenna keep-out areas.
- **One antenna at each end, left and right**, as far apart as the board allows.
- **USB-C on the bottom edge**, with the battery and soil-sensor connectors on the top edge.
- **USB-C or two series-connected AA cells may power the board.** USB takes priority and must not feed current into the cells. Do not fit a charging circuit.
- **The complete 3.3 V rail is switched.** Use the timer and buck-boost wiring on sheet 3; do not leave the sensor or either radio on an unswitched supply.
- **ERC and DRC must report zero errors**, including schematic parity. This is a condition of delivery.

FABRICATION AND ASSEMBLY

- **Surface finish:** lead-free HASL.
- **Ground vias:** 0.3 mm finished hole, 0.6 mm pad. Stitch ground at 3 mm spacing or closer around both antenna keep-outs, along the 50 Ω feed and around the board edge. The DN038 helix inter-layer vias are separate antenna features: copy their 0.38 mm diameter from the TI Gerber and never connect them to GND.
- **Placement:** modules, connectors, switches and the converter on top. Small resistors and capacitors may be placed on the bottom.

Battery Power and Timed Power-off

TIMER	TPL5111 IC on the main PCB
CYCLE	15 minutes nominal, resistor-set
POWER OFF	ESP32 + LoRa + sensor
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TIMER REQUIREMENT

Fit TPL5111DDCR directly on the PCB. Fit R13 as 68.4 kΩ, 1% for a nominal 900-second cycle. Switch the complete 3V3_SWITCHED rail so the ESP32, LoRa module and sensor lose power together.

POWER AND TIMER WIRING

all nets and parts are on the main PCB

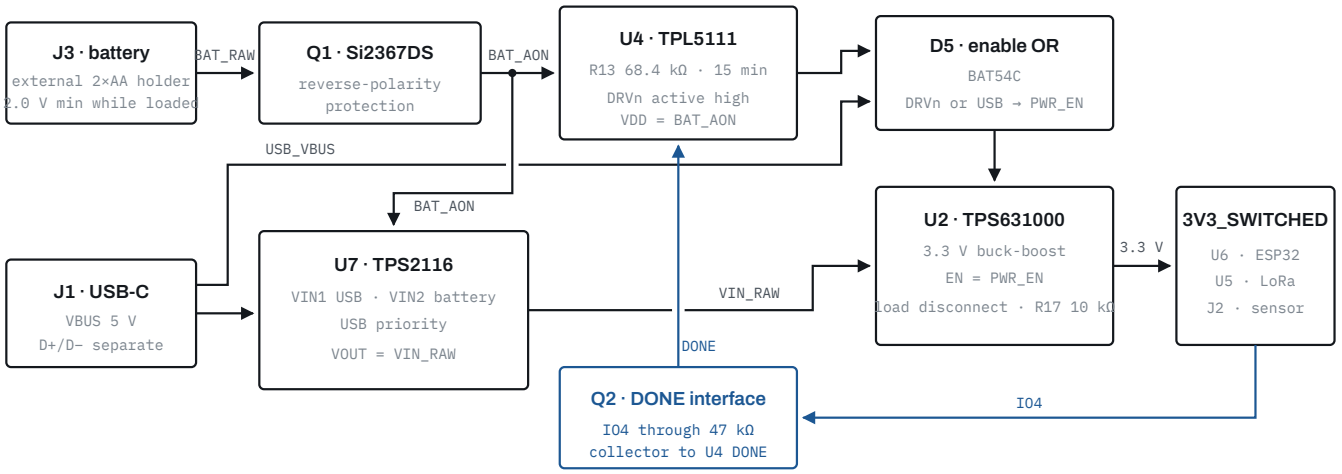


Fig 2 — Whole-unit power-off circuit. U4 is fitted on the main PCB. On battery power, U4 enables U2 every 15 minutes. U2's load-disconnect output is the only 3.3 V rail used by U6, U5 and J2, so the ESP32, LoRa module and soil sensor lose power together. USB VBUS overrides PWR_EN for uninterrupted programming and setup. U7 selects USB before battery and blocks reverse current; USB does not charge the battery.

EXACT TIMER CONNECTIONS

U4 PIN	CONNECT TO
1 · VDD	BAT_AON; C13 0.1 µF to GND at the pin
2 · GND	Ground
3 · DELAY	R13 68.4 kΩ, 1% to GND; SW3 momentary to BAT_AON
4 · DONE	Q2 collector; R12 100 kΩ pull-up to BAT_AON
5 · DRVn	D5 anode A; D5 common cathode is PWR_EN
6 · EN	Hard-wire to BAT_AON for repeating timer mode

Q2 is an MMBT3904 level shifter: pin 1 base from IO4 through R14 47 kΩ, pin 2 emitter to GND and pin 3 collector to U4 DONE; fit R15 1 MΩ base-to-GND. This prevents BAT_AON from reaching the 3.3 V GPIO and guarantees U4's 0.7 × VDD high threshold. **D5 is a BAT54C dual common-cathode diode:** pin 1 anode to U4 DRVn, pin 2 anode to USB VBUS and pin 3 common cathode to PWR_EN and R16 100 kΩ to GND. Thus U4 controls battery operation, while any USB connection holds the board on.

POWER INPUT AND CONVERTER

- **J3:** JST S2B-PH-SM4-TB, 2 pin, side entry. Pin 1 BAT_RAW, pin 2 GND. Fit on every board; leave empty when unused.
- **Battery:** external two-AA series holder; alkaline, NiMH or 1.5 V lithium primary. J3 range is 2.0 V minimum while loaded to 3.6 V maximum. Do not use a rectangular 9 V battery. Fit no charger.
- **Q1:** Vishay Si2367DS-T1-GE3 P-channel MOSFET for reverse-battery protection. Pin 3 drain to BAT_RAW, pin 2 source to BAT_AON, pin 1 gate to GND through R20 1 MΩ.
- **U7:** TPS2116DRLR power multiplexer. Pin 1 GND; pins 2 and 7 VIN_RAW; pin 3 USB_VBUS; pin 4 PR1; pin 5 MODE tied to USB_VBUS; pin 6 BAT_AON; pin 8 to a status test pad only. Make the PR1 divider with R21 330 kΩ from USB_VBUS to PR1 and R22 100 kΩ from PR1 to GND. Fit 1 µF to GND at VIN1, VIN2 and VOUT. This priority-mode wiring selects USB when present and blocks reverse current into either source.
- **U2:** TPS631000DRLR adjustable buck-boost converter. Pin 4 VIN to VIN_RAW; pin 1 VOUT to 3V3_SWITCHED; pin 5 EN to PWR_EN; pin 6 MODE to GND for power-save mode; pin 7 to GND. Connect a 1 µH inductor between pins 3 LX1 and 2 LX2. Set 3.3 V with R23 511 kΩ from VOUT to pin 8 FB and R24 91 kΩ from FB to GND. Fit 22 µF at VIN and 47 µF at VOUT, placed as shown in the TI reference layout and selected so their effective capacitance remains within the datasheet limits. Fit R17 10 kΩ from 3V3_SWITCHED to GND.
- **Power domains:** Q1 produces BAT_AON. Connect BAT_AON only to U4 VDD/EN, U7 VIN2 and the stated always-on pull resistors. U7 VOUT is VIN_RAW and connects to U2 VIN. U2 VOUT is 3V3_SWITCHED. Every ESP32, Wio-SX1262-N and sensor supply pin connects only to 3V3_SWITCHED.

Power-off Verification

OFF STATE	Complete 3.3 V power-off
RESUME TYPE	Cold boot
CONTROL PATH	IO4 through Q2 to DONE
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EXPECTED POWER STATES

INPUTS AND STATE	U4 DRVN	PWR_EN	3V3_SWITCHED	RESULT
Battery · active interval	high	high	3.3 V	ESP32, LoRa and sensor on
Battery · after DONE	low	low	<0.3 V after 5 s	ESP32, LoRa and sensor off
Battery + USB · after DONE	low	high	3.3 V	USB override keeps system on
No battery · USB connected	U4 off	high	3.3 V	Normal USB-powered operation

> **The 15 minutes is measured between timer wake edges.** If the active period takes 8 seconds, the power-off time is approximately 14 minutes 52 seconds. If DONE is never asserted, U4 keeps the board on until the end of the interval, removes power for about 50 ms, then starts the next cycle. With USB attached, D5 holds PWR_EN high, so a DONE edge does not switch off the board.

PHYSICAL CONTROL INTERFACE

IO4 LEVEL	Q2	U4 DONE PIN
high	on	low
low or high-Z	off	high

U4 switches off only when its DONE pin changes from low to high. Provide labelled test pads at IO4/Q2 base and U4 DONE so this edge can be checked without firmware. Do not connect IO4 directly to the always-on DONE net.

BRING-UP ACCEPTANCE

- Design and verify the power path for 550 mA peak at 3.3 V and approximately 1 A from a 2.0 V pack. Calculate input bulk capacitance from the measured load transient and AA cell, holder and contact resistance.
- On battery, verify a nominal 15-minute interval between power-up edges and document measured tolerance.
- Without USB, after DONE, verify 3V3_SWITCHED falls below 0.3 V within 5 s; confirm the ESP32, LoRa module and sensor are unpowered.
- Measure total battery current while powered off. Target **under 5 µA at 25 °C**, including input leakage and U2 shutdown current; record the actual result.
- Press SW3 while powered off and confirm a wake within 40 ms. With 5 V USB attached, confirm the board stays on and current from USB into J3 is at most 5 µA at 25 °C.
- Use a current-limited bench source at J3. Run a full sensor read and +22 dBm LoRa transmit while J3 remains at 2.0 V, then at 3.6 V. Capture droop at U2 VIN, U5 pad 8 and U6 pad 2; no brownout is allowed.
- Repeat the low-voltage transmit test with representative partly discharged AA cells, including holder and contact resistance. Record the practical battery cutoff.

Connections and Pin Allocation

RADIO PINS	Firmware-fixed SPI and control
SENSOR PORT	I²C, 3.3 V, always fitted
PCB	4 layers · 0.8 mm finished
BATTERY PORT	External 2×AA holder
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HOW THE PARTS CONNECT

blue is the sensor path

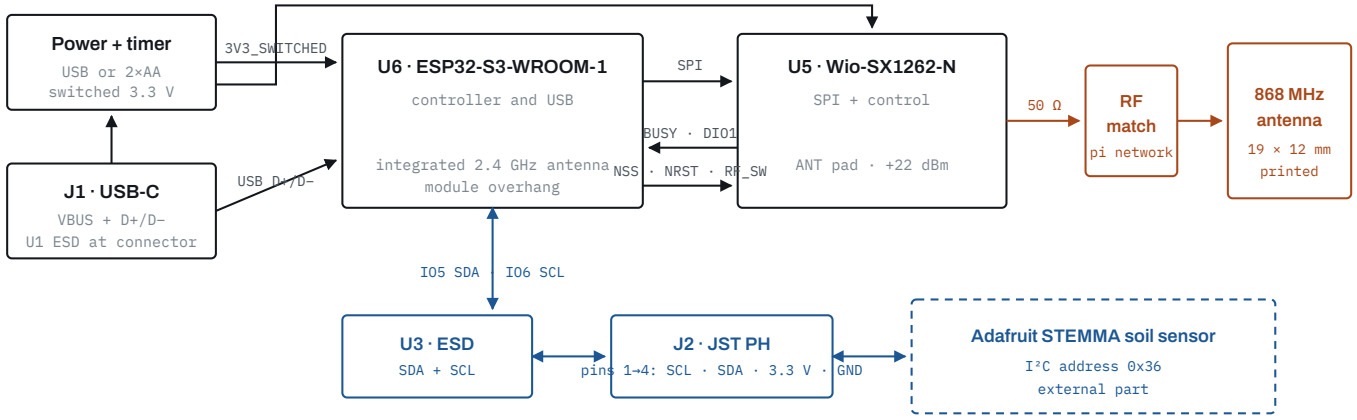


Fig 3 — **How the parts connect.** The battery holder, cells and soil sensor are off board. USB or the battery feeds 3V3_SWITCHED, which powers U6, U5 and J2. The sensor bus passes through U3 to J2. **J2 is 3.3 V only:** the sensor pulls SDA and SCL up to its VIN pin.

FIRMWARE-FIXED PINS

do not move these signals

SIGNAL	GPIO	ESP32 PAD	LORA PAD
SPI SCK	IO7	7	4
SPI MISO	IO8	12	2
SPI MOSI	IO9	17	3
NSS	IO41	34	6
BUSY	IO40	33	11
DIO1	IO39	32	12
NRST	IO42	35	5
USB D−	IO19	13	–
USB D+	IO20	14	–
User LED, active low	IO48	25	–

USB D−/D+ are fixed by the ESP32-S3 silicon. The other rows are fixed by the existing application. Moving any row requires a software change.

THE SENSOR CONNECTOR

J2 · pin order is fixed by the sensor

PIN	SIGNAL	GOES TO
1	**SCL**	Recommended IO6, through U3
2	**SDA**	Recommended IO5, through U3
3	**VIN**	**3V3_SWITCHED.** Not 5 V
4	**GND**	Ground

- **J2** — JST S4B-PH-SM4-TB, 4 pin, 2.0 mm pitch, side entry, surface mount. Put it on the top edge. Mark pin 1 on copper and silkscreen; include a numbered mating-face view on the schematic.
- **U3** — USBLC6-2P6 ESD protection on SDA and SCL, placed beside J2.
- **R10, R11** — 4.7 kΩ pull-up footprints on SDA and SCL. **Leave them empty.** The sensor already has 10 kΩ pull-ups. Fit only if the measured I²C rise time requires them; check the total sink current.
- **C12** — 1 µF capacitor on the 3.3 V feed to the connector.
- The sensor answers at I²C address 0x36.
- Fit J2. Leave it empty when no sensor is connected.

Recommended new connections

SIGNAL	GPIO	CHECK BEFORE RELEASE
Timer DONE	IO4	Check output use, strapping and module conflicts
I²C SDA	IO5	Check available on N8R8 and suitable for I²C
I²C SCL	IO6	Check available on N8R8 and suitable for I²C
RF_SW	IO38	U5 pad 1; 10 kΩ pull-up. RX: H/DIO2 L; TX: L/DIO2 H; off: both L

These GPIOs may move if a datasheet conflict is found; report any change. Pull IO0 up to 3.3 V with 10 kΩ; SW1 connects IO0 to GND. SW2 connects EN to GND. Verify all N8R8 strapping pulls.

HARDWARE DESIGN BRIEF

Parts and Deliverables

PCB	4 layers · 0.8 mm finished
ASSEMBLY	BOM and placement CSV
CONNECTORS	Battery and I²C always fitted
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MAIN PARTS

REF	REQUIRED PART
U6	ESP32-S3-WROOM-1-N8R8 — 8 MB flash, 8 MB PSRAM, integrated antenna
U5	Seeed Wio-SX1262-N without IPEX, SKU 113991436. Do not substitute the IPEX-equipped version; its ANT pad is not connected
U2	TPS631000DRLR — adjustable buck-boost converter with load disconnect
U4	TPL5111DDCR — system timer, nominal 15-minute cycle
U7	TPS2116DRLR — USB-priority power multiplexer
Q1	Vishay Si2367DS-T1-GE3 — P-channel MOSFET specified at VGS = −1.8 V, SOT-23
Q2	onsemi MMBT3904LT1G — NPN transistor, SOT-23
D5	Diodes Inc. BAT54C-7-F — dual common-cathode Schottky, SOT-23
U1, U3	USBLC6-2P6 — ESD protection, SOT-666
J1	HRO TYPE-C-31-M-12 — 16-contact USB 2.0 Type-C receptacle. Lock the procurement source and use the HRO Rev A manufacturer drawing; do not use a footprint from an unverified clone
J2	JST S4B-PH-SM4-TB — 4-pin side-entry sensor connector
J3	JST S2B-PH-SM4-TB — 2-pin connection to external 2×AA holder
J4	Hirose U.FL-R-SMT-1(10) — VNA test connector, DNP
SW1–SW3	Omron B3U-1000P — BOOT, RESET and manual timer trigger
R13	68.4 kΩ, 1% — timer interval
R17	10 kΩ — switched-rail discharge
R18, R19	0 Ω RF jumpers — production fits R18 only; VNA setup fits R19 only
L1	Murata DFE252012P-1R0M=P2 — 1 µH, 4.3 A saturation current
Off board	Adafruit STEMMA soil sensor and 4-pin JST PH cable

WHAT TO DELIVER

- **KiCad project** with separate sheets for power/timer, microcontroller, LoRa/radio and the I²C plug.
- **Schematic PDF.** Battery and I²C connectors and the timer are part of the design.
- **PCB layout** with zero ERC and DRC errors, including schematic parity, plus Gerber and drill files.
- **Part list and placement file** in the factory's CSV format. J1, J2 and J3 are fitted; the VNA-only J4 footprint and R19 are DNP.
- **Fabrication notes** naming four layers, 0.8 mm finished thickness, the released impedance stack-up, surface finish, via size and assembly sides.
- **Antenna tuning record** with the final matching values and measured VSWR across 863–870 MHz.
- **3D STEP file** for enclosure design.

REQUIRED SUPPORT WIRING

verify against the released manufacturer documents

- **U5:** pads 8 to 3V3_SWITCHED, 7 and 10 to GND, and 9 to the 50 Ω RF path. Fit 100 nF and 10 µF beside pad 8. Provide series-damping footprints in SCK, MOSI, MISO and NSS; fit 22 Ω initially and verify the values during bring-up.
- **U6:** pad 2 to 3V3_SWITCHED; pads 1, 40 and exposed pad 41 to GND. Use Espressif's reference decoupling. Pull EN up with 10 kΩ and fit 1 µF from EN to GND. Verify against the N8R8 module guidance.
- **J1:** use the exact HRO mating-face names. Join A6+B6 as D+ and A7+B7 as D−. Join A4+A9+B4+B9 as USB_VBUS; join A1+A12+B1+B12 to GND. Fit separate 5.1 kΩ resistors from A5/CC1 and B5/CC2 to GND. Leave A8/B8 SBU unconnected. Connect all four shell tabs directly to board GND at the connector with short copper and multiple vias.
- **USB data:** route D−/D+ as a 90 Ω differential pair through U1, then through series footprints at U6 to IO19/IO20. Fit 0 Ω initially; change to 22–33 Ω only after signal-integrity testing. Optional shunt capacitors are DNP. The design is a USB 2.0 sink with no PD controller; do not assume more than 500 mA from VBUS, and verify operation from a 5 V source current-limited to 500 mA.
- **U1 and U3:** follow the USBLC6-2P6 flow-through pinout. Pin 2 is GND. Pin 5 connects to USB_VBUS for U1 and 3V3_SWITCHED for U3. Place each device beside its connector and fit 100 nF directly between pins 5 and 2.
- **User LED:** connect the LED anode to 3V3_SWITCHED through 1 kΩ and its cathode to IO48. This makes the firmware-fixed LED active low.
- **Release checks:** verify J3 polarity and mark it on copper and silkscreen. Keep the U2 switch node away from both antennas. Apply the antenna keep-outs on sheet 2 and tune the assembled 868 MHz antenna.